

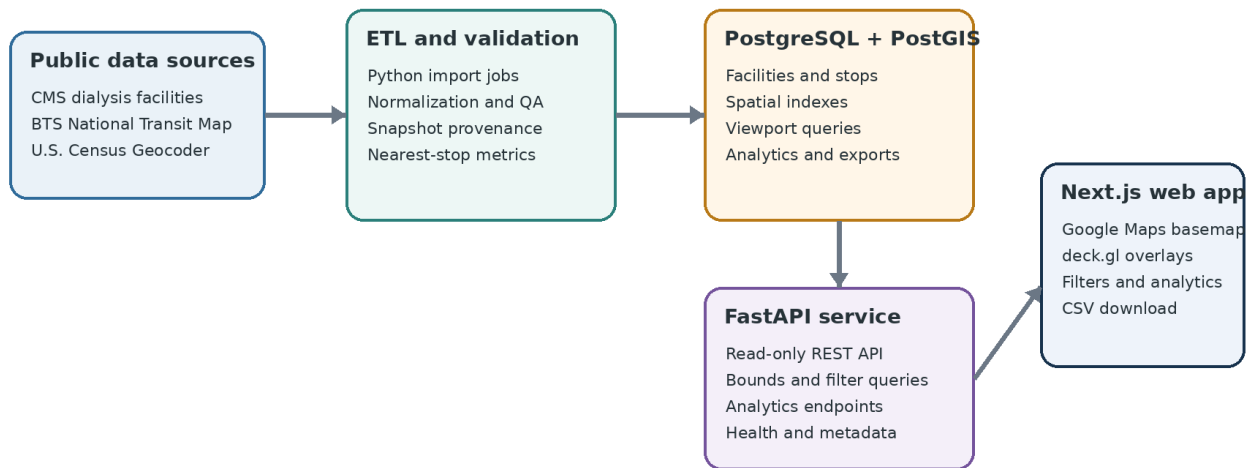
Dialysis & Transit Explorer

Phase 1 Product and Technical Design Specification

Version	Version 1.0
Date	July 29, 2026
Audience	Codex and implementation reviewers
Status	Authoritative Phase 1 build specification

Build target: a public, read-only national geospatial explorer combining CMS dialysis facilities and BTS transit stops on a Google Maps basemap.

Phase 1 Reference Architecture



Scientific framing: Phase 1 measures geographic proximity only; it does not establish functional transit accessibility.

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1. Document purpose

This document is the authoritative build specification for the Phase 1 version of the Dialysis & Transit Explorer, a public-facing exploratory web application that overlays Medicare-certified dialysis facilities and public transportation stops on a Google Maps basemap and lets users perform reproducible proximity analytics.

Codex should treat this document as the product requirements document, system design, implementation plan, and acceptance-test specification. When details are ambiguous, Codex should favor the simplest implementation that satisfies the stated acceptance criteria and should not silently expand the Phase 1 scope.

2. Product summary

The Phase 1 product is a read-only national map and analytics tool. It combines:

- Dialysis facility records from the CMS Provider Data Catalog.
- Fixed-route transit stops from the Bureau of Transportation Statistics National Transit Map.
- Stored geocodes for dialysis facilities produced by the U.S. Census Geocoder.
- A Google Maps JavaScript basemap.
- Custom WebGL point layers rendered with deck.gl.
- PostGIS calculations for nearest-stop distance, stop counts within user-selected radii, filtering, grouping, and CSV export.

The application is exploratory. It measures geographic proximity, not route availability, schedule compatibility, pedestrian accessibility, paratransit eligibility, or the ability of an individual patient to complete a trip.

3. Goals

1. Display dialysis facilities and public transportation stops throughout the United States without attempting to load the full national stop dataset into the browser.
2. Let users navigate to a state, city, ZIP code, address, or map extent and inspect local dialysis-transit relationships.
3. Let users select a straight-line distance threshold and calculate how many facilities have at least one transit stop within that distance.
4. Let users inspect any dialysis facility and view its nearest stop, nearest-stop distance, and number of stops within the selected threshold.
5. Let users compare filtered regions and facility characteristics using summary cards, a distance distribution, and a sortable facility table.
6. Let users download the currently filtered facility-level results as CSV.
7. Preserve source names, snapshot dates, import status, geocoding status, and limitations so results can be interpreted and reproduced.
8. Provide a maintainable base for later schedule-aware GTFS, walking-network, Census-demographic, and patient-journey analyses.

4. Phase 1 non-goals

- Real-time transit positions or GTFS-Realtime.
- Transit trip planning, arrival prediction, or navigation.
- Route, trip, stop-time, calendar, fare, or service-frequency analytics.
- Walking-network or driving-network distances.
- Dialysis shift schedules or treatment-time compatibility.
- Census demographic or social-vulnerability overlays.
- Patient-level data, protected health information, clinical predictions, or missed-treatment modeling.
- User accounts, saved projects, collaboration, comments, or permissions.

- Facility editing by end users.
- Native mobile applications.
- Google Places as the authoritative source of dialysis facilities or transit stops.
- A claim that stop proximity proves practical accessibility.

5. Primary users and use cases

5.1 Users

- Health-services and transportation researchers.
- Dialysis program administrators.
- Transit planners and municipal staff.
- Public-health students and educators.
- Community advocates.
- Clinicians exploring transportation context.

5.2 Core use cases

1. **National overview:** View the geographic distribution of dialysis facilities and transit stops.
2. **State comparison:** Select a state and calculate the percentage of dialysis facilities within 400 meters of a transit stop.
3. **Local exploration:** Search for Tucson, Arizona, zoom to the area, and inspect individual facilities and nearby stops.
4. **Threshold sensitivity:** Change the threshold from 400 meters to 800 meters and see the summary, chart, and table update.
5. **Facility review:** Select a facility and inspect source attributes, geocoding quality, nearest stop, and nearby-stop count.
6. **Gap identification:** Sort facilities from greatest to least nearest-stop distance.
7. **Data export:** Download the filtered facility metrics with source snapshot metadata.

6. Data sources and provenance

6.1 Dialysis facilities

Source: CMS Provider Data Catalog, dataset “Dialysis Facility - Listing by Facility.” Dataset identifier: 23ew-n7w9. The CMS Certification Number (CCN) is the authoritative facility identifier and must be preserved as text, including leading zeroes.

Minimum imported fields:

- CMS Certification Number (CCN)
- Facility name
- Address lines 1 and 2
- City/town, state, ZIP code, and county/parish
- Telephone number
- Profit or non-profit
- Chain owned and chain organization
- Late shift
- Number of dialysis stations
- In-center hemodialysis, peritoneal dialysis, and home hemodialysis training indicators
- Certification date
- CMS source release or snapshot date

The facility listing does not provide a complete durable coordinate field for this application, so Phase 1 geocodes addresses during ETL and records the geocoder match status.

6.2 Dialysis facility geocoding

Primary geocoder: U.S. Census Geocoder using the current public address-range benchmark.

- Submit records in batches of no more than 10,000.
- Preserve the original CMS address exactly.
- Store matched address, longitude, latitude, match indicator, match type, and benchmark.
- Use geocode statuses exact, non_exact, tie, no_match, and manual_override.
- Do not silently discard unmatched facilities.
- Produce an import QA report listing unmatched and tied facilities.
- Permit a version-controlled override CSV for verified coordinates.

Census geocodes are interpolated along address ranges and should be labeled as approximate.

6.3 Transit stops

Source: Bureau of Transportation Statistics, National Transit Map Stops. The current implementation target is the public ArcGIS FeatureServer listed below.

```
https://services.arcgis.com/x0ilkZaI0eWDREZv/arcgis/rest/services/  
NTAD_National_Transit_Map_Stops/FeatureServer/0
```

Minimum imported fields:

- OBJECTID, ntd_id, feed_id, stop_id, stop_name, and stop_desc
- stop_lat and stop_lon
- stop_url and stop_code
- location_type and parent_station
- stop_timezone
- wheelchair_boarding
- platform_code
- stop_type and stop_type_text
- download_date and source snapshot date

The importer must paginate because the ArcGIS service has a record limit. Query only required fields and geometry. Store the source object ID and a deterministic compound key such as feed_id + stop_id + location_type when available.

6.4 Data attribution and limitations

- CMS data represent Medicare-certified dialysis facilities and may include facilities that are closed, temporarily inactive, specialized, or not accepting new patients unless the source explicitly indicates otherwise.
- National Transit Map coverage and source validity vary by agency and snapshot.
- The National Transit Map supports research and analysis and is not a navigation or real-time trip-planning product.
- Straight-line proximity does not measure route schedules, walking barriers, safety, weather exposure, disability access, or trip feasibility.
- Facility coordinates are approximate when derived from address-range geocoding.

Display attribution for CMS, the U.S. Census Bureau, USDOT/BTS, and Google Maps. Preserve Google Maps attribution without alteration.

7. Product behavior

7.1 Default state

- Center the contiguous U.S. near latitude 39.5, longitude -98.35 at zoom 4.
- Display dialysis-facility clusters.
- Display transit-stop aggregate clusters, not raw stops.
- Set the proximity threshold to 400 meters.
- Use all facility ownership and service filters, with in-center hemodialysis set to yes.
- Show a brief methods notice and link to the full Data & Methods panel.

7.2 Map layers

Layer controls:

- Dialysis facilities: on by default.
- Transit stops: on by default.
- Nearest-stop connection line: off by default; available for a selected facility.
- Facility threshold circle: off by default; available for a selected facility.

Rendering rules:

- Use Google Maps JavaScript API for the basemap and map controls.
- Use deck.gl WebGL layers for raw points and server-generated clusters.
- Do not render hundreds of thousands of HTML markers.
- Use distinct shape, size, and color encoding for facilities and stops; do not rely on color alone.
- Return raw transit stops only when zoom and viewport produce a manageable count.
- At lower zoom levels, return aggregates with counts and centroids.

7.3 Navigation and search

- Google Maps location autocomplete for address/city/ZIP navigation, with no persistence of Google Places content beyond the active interaction.
- State dropdown.
- Reset-to-national button.
- Current-location button only after explicit browser permission.
- URL parameters for map center, zoom, state, radius, and primary filters.

7.4 Filters

Facility filters:

- State.
- In-center hemodialysis offered: default yes.
- Ownership: all, for-profit, non-profit, government/other as represented in source.
- Chain owned: all, yes, no.
- Chain organization: multi-select populated from current data.
- Late shift: all, yes, no, unknown.
- Minimum and maximum number of dialysis stations.
- Geocoding status: matched only by default; unresolved records may appear in tables but not spatial analytics.

Transit filters:

- stop_type_text multi-select when populated.
- Wheelchair boarding status: all, indicated accessible, indicated not accessible, unknown.
- Agency/NTD ID multi-select, searchable and limited to the current map or state.

7.5 Analysis extent

Phase 1 supports two analysis extents:

1. Selected state when a state filter is active.
2. Current map viewport when no state filter is active or when the user explicitly chooses “Analyze current map.”

The UI must label the active extent clearly. Do not imply that a viewport summary is a formal administrative-area statistic.

7.6 Proximity threshold

Provide preset thresholds of 250, 400, 800, and 1,600 meters, plus a custom threshold from 100 to 5,000 meters.

All Phase 1 distances are geodesic straight-line distances calculated in PostGIS using geography types. Display values in meters below 1,000 meters and miles at or above 1,000 meters, while retaining metric values in details and CSV.

7.7 Analytics panel

For the active extent and filters, display:

- Dialysis facilities included.
- Facilities with at least one transit stop within threshold.
- Percentage with a stop within threshold.
- Median nearest-stop distance.
- 25th and 75th percentile nearest-stop distance.
- Transit stops in the active extent.
- Facilities with unresolved geocodes excluded from spatial analytics.

Distance-distribution bins:

Bin	Definition
0-250 m	Nearest stop at or below 250 m
251-400 m	More than 250 m through 400 m
401-800 m	More than 400 m through 800 m
801-1,600 m	More than 800 m through 1,600 m
More than 1,600 m	More than 1,600 m
No valid geocode	Spatial metric not calculated

The sortable facility table contains facility name, city/state, ownership, chain organization, station count, nearest stop name, nearest-stop distance, stops within threshold, geocoding status, and CCN. Default order is greatest nearest-stop distance first, with unresolved geocodes last.

7.8 Facility detail drawer

- Facility name, address, CCN, and telephone number.
- Ownership, chain, station count, modalities, and late-shift field.
- Geocoding status, matched address, and coordinate source.
- Nearest-stop identity, type, agency, distance, and wheelchair field.
- Stop count within the active threshold.
- Controls for threshold circle and nearest-stop connection line.
- Source snapshot dates and limitations notice.

Selecting a transit stop opens a smaller detail card with source identifiers, name, type, agency, wheelchair field, URL if present, and snapshot date.

7.9 CSV export

Export one row per filtered dialysis facility and include all table fields, facility coordinates, geocoding metadata, nearest-stop identifiers and coordinates, nearest-stop distance, stop count, threshold, extent description, source snapshot dates, and export time in UTC.

```
dialysis-transit-explorer_YYYY-MM-DD_<state-or-viewport>_<radius-m>.csv
```

8. System architecture

8.1 Required stack

Layer	Technology
Frontend	Next.js with TypeScript and App Router
Map	Google Maps JavaScript API with vector map ID
Overlay rendering	deck.gl through @deck.gl/google-maps
Charts	Accessible React chart library such as Recharts, with text/table equivalents
Backend	FastAPI with Pydantic and SQLAlchemy 2.x
Database	PostgreSQL with PostGIS
ETL	Python using httpx, pandas, psycopg or SQLAlchemy
Local orchestration	Docker Compose
Packages	pnpm for web; locked Python dependencies with uv or equivalent
Testing	Vitest, React Testing Library, Playwright, and Pytest

Use current stable releases at implementation time and commit lockfiles. Avoid preview or experimental framework features unless the specification cannot be met otherwise.

8.2 Component responsibilities

Next.js web application

- Renders the UI.
- Owns client-side map and filter state.
- Calls FastAPI.
- Does not connect directly to PostgreSQL.
- Does not expose server credentials.

FastAPI service

- Provides read-only spatial and metadata APIs.
- Validates bounds, filters, radii, pagination, and sorting.
- Returns clusters or raw points based on zoom.
- Runs analytics and streams CSV exports.
- Exposes health and snapshot metadata endpoints.

PostgreSQL/PostGIS

- Stores normalized source data and metadata.
- Stores geometries in EPSG:4326.
- Uses geography for meter-based distance.
- Maintains spatial and filter indexes.
- Stores precomputed nearest-stop metrics.

ETL jobs

- Fetch and snapshot source data.
- Validate schemas.
- Transform and load staging tables.
- Geocode addresses and apply overrides.
- Swap staging data transactionally.
- Compute metrics and QA reports.

9. Repository layout

```
/
├── apps/
│   └── web/                                # Next.js application
├── services/
│   └── api/                                # FastAPI application
├── pipelines/
│   ├── fetch_cms.py
│   ├── fetch_ntm_stops.py
│   ├── geocode_facilities.py
│   ├── load_database.py
│   ├── compute_metrics.py
│   └── qa_report.py
├── data/
│   ├── raw/                               # gitignored immutable snapshots
│   ├── processed/                         # gitignored transformed artifacts
│   └── overrides/
│       └── facility_coordinates.csv
├── db/
│   ├── migrations/
│   └── seeds/
├── infra/
│   ├── docker-compose.yml
│   └── Dockerfiles/
├── docs/
│   ├── DESIGN.md
│   ├── DATA_DICTIONARY.md
│   ├── METHODS.md
│   └── OPERATIONS.md
├── tests/
│   └── fixtures/
├── .env.example
├── Makefile
└── README.md
```

10. Database design

10.1 source_snapshots

Column	Type	Notes
id	UUID	Primary key
source_name	text	cms_dialysis, bts_ntm_stops, or census_geocoder
source_dataset_id	text	CMS dataset ID or ArcGIS item/service ID
source_url	text	Retrieval endpoint
source_published_at	timestampz nullable	Source-provided date
retrieved_at	timestampz	Required
content_sha256	text	Raw artifact hash
row_count	bigint	Loaded source rows
status	text	started, validated, loaded, or failed
metadata	jsonb	Schema/version details

10.2 dialysis_facilities

Column	Type	Notes
ccn	text	Primary key; preserve leading zeroes
facility_name	text	Required
address fields	text	Address lines, city, state, ZIP, county
telephone	text nullable	Source field
profit_status	text nullable	Normalized plus raw value
chain_owned	boolean nullable	Source field
chain_organization	text nullable	Indexed
late_shift	boolean nullable	Source field
dialysis_stations	integer nullable	Source field
modality flags	boolean nullable	In-center HD, PD, home-HD training
certification_date	date nullable	Source field
source_snapshot_id	UUID	Foreign key
source_record	jsonb	Selected original fields
geom	geometry(Point,4326)	Nullable; GiST index
geocode_status	text	Indexed
geocode_match_type	text nullable	Match detail
geocode_matched_address	text nullable	Matched address

Column	Type	Notes
geocode_benchmark	text nullable	Census benchmark
geocode_source	text nullable	census or manual_override
updated_at	timestamptz	Audit time

10.3 transit_stops

Column	Type	Notes
id	bigserial	Internal primary key
source_object_id	bigint	Indexed
ntd_id / feed_id	text nullable	Indexed
stop_id / stop_name	text nullable	Source fields
stop_desc / stop_url	text nullable	Source fields
stop_code	text nullable	Source field
location_type	text nullable	Source field
parent_station	text nullable	Source field
stop_timezone	text nullable	Source field
wheelchair_boarding	text nullable	Indexed
platform_code	text nullable	Source field
stop_type / stop_type_text	text nullable	Indexed
download_date	date nullable	Source field
source_snapshot_id	UUID	Foreign key
geom	geometry(Point,4326)	Required; GiST index
dedupe_key	text	Unique within snapshot

Do not collapse stops simply because coordinates match; paired directional stops may legitimately share or nearly share coordinates. Deduplicate only exact duplicate source records using deterministic source identifiers.

10.4 facility_stop_metrics

Column	Type	Notes
facility_ccn	text	Primary and foreign key
nearest_stop_pk	bigint nullable	Foreign key
nearest_stop_distance_m	double precision nullable	Geography distance
stops_within_250m	integer	Preset count
stops_within_400m	integer	Preset count
stops_within_800m	integer	Preset count
stops_within_1600m	integer	Preset count

Column	Type	Notes
cms_snapshot_id	UUID	Source lineage
ntm_snapshot_id	UUID	Source lineage
computed_at	timestampz	Audit time

Calculate custom-radius stop counts on demand. Preset columns support fast default analytics.

11. ETL specification

11.1 General rules

- Create a source_snapshots row before processing.
- Save immutable raw responses under a timestamped snapshot directory.
- Validate required columns before loading.
- Abort production swap if required fields are missing or row counts are implausibly low.
- Load staging tables and swap transactionally.
- Retain the previous successful snapshot for rollback.
- Log counts, durations, errors, and schema changes.
- Never modify raw source files.

11.2 CMS import

Use the Provider Data Catalog API endpoint pattern documented by CMS:

```
https://data.cms.gov/provider-data/api/1/datastore/query/23ew-n7w9/0
```

- Paginate until all rows are retrieved.
- Preserve CCN as text.
- Normalize yes/no fields to nullable booleans while retaining raw values.
- Normalize state abbreviations to uppercase.
- Validate CCN uniqueness.
- Compare row count and state coverage with the previous snapshot.

11.3 Census geocoding

- Geocode only new or changed addresses and records lacking an override.
- Use batch processing where possible.
- Cache results by normalized-address hash.
- Apply data/overrides/facility_coordinates.csv after automated geocoding.
- Use override columns ccn, latitude, longitude, verified_address, verification_source, verified_at, and notes.
- Produce exact, non-exact, tie, no-match, and override counts.

11.4 NTM stop import

Use the ArcGIS query endpoint and these recommended parameters:

```
where=1=1
outFields=OBJECTID,ntd_id,feed_id,stop_id,stop_name,stop_desc,
stop_lat,stop_lon,stop_url,stop_code,location_type,parent_station,
stop_timezone,wheelchair_boarding,platform_code,stop_type,
stop_type_text,download_date
```

```
returnGeometry=true
outSR=4326
f=geojson
resultOffset=<offset>
resultRecordCount=<page size>
```

- Discover the service maximum record count from metadata.
- Paginate until no records remain.
- Retry transient failures with exponential backoff.
- Do not remove valid Alaska, Hawaii, Puerto Rico, or island-area records.
- Record total features and null-field rates.

11.5 Metric computation

- For each geocoded facility, find the nearest stop using K-nearest-neighbor ordering and confirm exact geography distance.
- Count stops within 250, 400, 800, and 1,600 meters.
- Compute in batches using spatial indexes.
- Store source snapshot IDs.
- Treat unresolved facility metrics as not calculated, not as no nearby stops.

12. API design

Base path: /api/v1. All responses include a request ID. Errors use a consistent structure:

```
{
  "error": {
    "code": "INVALID_RADIUS",
    "message": "Radius must be between 100 and 5000 meters.",
    "request_id": "..."
  }
}
```

12.1 Health and metadata

- GET /health/live
- GET /health/ready
- GET /metadata/snapshots
- GET /metadata/filters

The filter metadata endpoint returns available states, ownership values, chain organizations, stop types, NTD IDs, and source snapshot dates.

12.2 Map endpoints

GET /map/facilities

Accept bounds, zoom, facility filters, and limit. Return clusters below the raw-point zoom and raw facilities when the viewport count is manageable. Include a cluster-or-point mode field.

GET /map/stops

Accept bounds, zoom, stop filters, and limit. Return server-generated clusters at low and medium zooms. Return raw points only when the expected count is below the safe threshold. Otherwise return coarser aggregation with `resolution_limited=true`.

Zoom	Mode	Initial precision
0-5	aggregate	3
6-7	aggregate	4
8-10	aggregate	5
11-12	aggregate	6
13+	raw if safe; otherwise aggregate	7

Tune precision using performance tests.

12.3 Facility detail

- GET /facilities/{ccn}
- GET /facilities/{ccn}/nearby-stops?radius_m=400&limit=100

12.4 Analytics

GET /analytics/summary

Accept state or viewport extent, radius, facility filters, and stop filters. Return the Section 7.7 metrics and distribution-bin counts.

GET /analytics/facilities

Return paginated facility-level metrics with allowlisted sort fields and a maximum page size of 500.

GET /analytics/export.csv

Use the same filters and stream CSV rather than constructing the full file in browser memory.

12.5 Parameter validation

- Bounds must be numeric and ordered.
- Radius must be 100-5,000 meters.
- Zoom must be 0-22.
- Page size must respect endpoint maximums.
- Map public sort/filter names to an allowlist; never interpolate arbitrary SQL field names.
- Apply request and database statement timeouts.

13. Spatial query rules

- Store points as geometry(Point,4326).
- Use bounding-box comparisons for viewport prefiltering.
- Cast to geography for ST_DWithin and ST_Distance.
- Use ST_MakeEnvelope for ordinary viewports.
- Handle antimeridian-crossing bounds explicitly.
- Do not use Web Mercator planar distance for reported metrics.
- Round only for display; preserve full precision in database and export.

14. Frontend design

14.1 Desktop layout

- Top application bar, 56-64 px high.
- Collapsible left panel, approximately 340 px wide.
- Full-height map in remaining space.
- Right-side facility detail drawer, approximately 380 px wide.
- Floating legend and controls that do not obscure Google attribution.

Left-panel tabs: Layers, Filters, Analytics, and Data & Methods.

14.2 Mobile layout

- Full-screen map.
- Bottom sheet for layers, filters, analytics, and details.
- Search and state controls remain available.
- Minimum touch target of 44 by 44 CSS pixels.
- Use cards rather than the full table on narrow screens; retain CSV export.

14.3 Visual language

- Use a clean research-tool aesthetic.
- Use distinct non-branded symbols for facilities and transit stops.
- Show counts in clusters.
- Increase size and add a halo to selected features.
- Use a colorblind-safe palette with shape or outline redundancy.
- Display data freshness in the visible interface.

14.4 Loading, empty, and error states

- Use skeletons or progress indicators for analytics and tables.
- Keep the map interactive while analytics load.
- Cancel superseded requests.
- Debounce viewport requests by approximately 250-400 ms.
- Explain empty results caused by filters, zoom, unresolved geocodes, or missing source coverage.
- Preserve the last successful map state on API failure and offer retry.

15. Accessibility

Target WCAG 2.2 AA.

- Keyboard-accessible controls and visible focus styles.
- Programmatic labels for all controls.
- Screen-reader announcements for layer state and selected feature.
- Text and table equivalents for charts.
- No color-only encoding.
- Table-based alternative to map exploration.
- Respect reduced-motion settings.
- Do not trap keyboard focus in the map.
- Provide an accessibility statement and feedback-contact placeholder.

16. Security and privacy

- No patient data or user accounts.
- Do not persist user-entered addresses in application logs.
- Restrict the Google Maps browser API key by HTTP referrer and required APIs.
- Keep database credentials and server configuration in environment variables.
- Use read-only database credentials for the public API.
- Apply production CORS allowlists.
- Set CSP, HSTS, X-Content-Type-Options, and a suitable referrer policy.
- Rate-limit expensive analytics and exports.
- Prevent spreadsheet-formula injection in CSV.
- Run dependency and secret scanning in CI.
- Publish Terms of Use and Privacy Policy pages compatible with Google Maps Platform requirements.

17. Performance requirements

Target	Requirement
Initial application shell	Interactive within 3 seconds on typical broadband
National facility clusters	p95 under 1.5 seconds
National stop aggregates	p95 under 2 seconds
State summary, preset radius	p95 under 2 seconds
Facility detail	p95 under 500 ms
Main-thread blocking	No more than 100 ms during map interactions
Raw stop response	Configurable cap, initially 20,000 points
Response size	Generally under 5 MB compressed

Use spatial indexes, preset metrics, server aggregation, compression, query caching, request cancellation, and client-side deduplication.

18. Observability and operations

- Structured JSON logs with request IDs.
- Metrics for request count, latency, status, query time, response size, and cache hit rate.
- ETL metrics for row counts, geocode rates, failures, and duration.
- Health endpoints verify process and database connectivity.
- Display successful source snapshot dates in the UI.
- Document refresh, rollback, and coordinate-override procedures.
- Never replace production data with a failed refresh.

19. Testing strategy

19.1 Unit tests

- Source normalization.
- Address hashing and override application.
- Radius and bounds validation.

- Distance formatting.
- Filter-to-URL serialization.
- CSV formula-injection protection.
- Cluster precision selection.

19.2 Database tests

- Nearest-stop selection.
- Exact meter distances within tolerance.
- Stop counts at boundary distances.
- Viewport inclusion.
- State and facility filters.
- Null geocode handling.
- Antimeridian logic.

19.3 API integration tests

- Valid and invalid parameters.
- Cluster and raw modes.
- Pagination and sorting.
- Summary/table population consistency.
- CSV column order and metadata.
- Snapshot metadata.
- Timeout and rate-limit behavior.

19.4 Frontend tests

- Layer toggles.
- Filter updates.
- Threshold changes.
- Facility drawer.
- URL state restoration.
- Empty and error states.
- Keyboard navigation and screen-reader labels.

19.5 End-to-end tests

1. Load national view and confirm cluster layers.
2. Select Arizona and obtain analytics.
3. Search Tucson and zoom to the city.
4. Select a facility and show nearest-stop line.
5. Change radius from 400 to 800 meters and verify updates.
6. Sort table by nearest-stop distance.
7. Export CSV and verify headers and data.
8. Reload a shared URL and restore state.

20. Local development and deployment

20.1 Environment variables

NEXT_PUBLIC_GOOGLE_MAPS_API_KEY=

```
NEXT_PUBLIC_GOOGLE_MAP_ID=
NEXT_PUBLIC_API_BASE_URL=http://localhost:8000/api/v1
DATABASE_URL=postgresql+psycopg://...
PUBLIC_DATABASE_URL=postgresql+psycopg://...read_only...
CMS_DATASET_ID=23ew-n7w9
CMS_DATASTORE_RESOURCE_INDEX=0
BTS_NTM_STOPS_FEATURESERVER=https://services.arcgis.com/x0ilkZaI0eWDREZv/arcgis/rest/services/
NTAD_National_Transit_Map_Stops/FeatureServer/0
DATA_SNAPSHOT_DIR=/data/raw
APP_ENV=development
API_RATE_LIMIT_PER_MINUTE=120
RAW_POINT_LIMIT=20000
```

20.2 Required commands

```
make bootstrap      # install dependencies and start database
make migrate        # apply migrations
make ingest-sample  # load fixture data
make ingest-production # fetch and load complete sources
make compute-metrics
make dev            # start web and API
make test
make lint
make e2e
make qa-report
```

20.3 Deployment model

- Deploy Next.js and FastAPI as separate services.
- Use managed PostgreSQL with PostGIS enabled.
- Run ETL as a controlled scheduled job, initially manual or quarterly.
- Use development, staging, and production environments.
- Do not expose the database publicly.
- Deploy only after staging completes the smoke-test checklist.

21. Phase 1 acceptance criteria

Data

- CMS import completes with unique CCNs.
- Every facility has an explicit geocoding status.
- NTM import paginates through the full snapshot.
- Source hashes, row counts, and dates are recorded.
- A QA report is generated.
- Nearest-stop metrics are computed for all geocoded facilities.

Map

- National layers render without loading the full stop dataset in browser.
- Clusters transition to raw points appropriately.
- Map, state, and URL navigation work.
- Feature selections show correct details.
- Google Maps attribution remains visible.

Analytics

- Summary responds to extent, filters, and radius.
- Table population matches the summary query population.
- Fixture facility metrics match expected values.
- Preset and custom radii work from 100 to 5,000 meters.
- CSV contains required fields and metadata.

Quality

- Unit, integration, and end-to-end tests pass.
- No unaddressed high-severity dependency vulnerabilities.
- Basic WCAG 2.2 AA audit passes for non-map controls and table alternatives.
- Production build succeeds from a clean checkout.
- README and operations documentation contain no undocumented required steps.

22. Implementation sequence for Codex

Epic 1: Repository and local platform

- Create monorepo structure.
- Configure Next.js, FastAPI, Postgres/PostGIS, Docker Compose, linting, formatting, and tests.
- Add health endpoints and a basic map shell.

Exit criterion: make bootstrap, make dev, and smoke tests work without source data.

Epic 2: Database and fixture data

- Add migrations and indexes.
- Add deterministic fixture data with known distances.
- Add database tests.

Exit criterion: fixture nearest-stop metrics pass exact expected values.

Epic 3: CMS ingestion and Census geocoding

- Implement CMS pagination and normalization.
- Implement Census batch geocoding and override workflow.
- Produce QA report.

Exit criterion: complete CMS snapshot loads with explicit status for every facility.

Epic 4: NTM stop ingestion

- Implement ArcGIS metadata discovery, pagination, retries, and staging load.
- Create indexes and QA summary.

Exit criterion: full NTM snapshot loads and row count matches source.

Epic 5: Metric computation and APIs

- Compute preset nearest-stop metrics.
- Implement metadata, map, detail, analytics, and export endpoints.
- Add validation, timeouts, and integration tests.

Exit criterion: fixture API tests pass and national queries are bounded.

Epic 6: Map interface

- Integrate Google Maps and deck.gl.
- Add layers, clustering, selections, state navigation, search, and URL state.

Exit criterion: national and Tucson exploration works without browser overload.

Epic 7: Filters and analytics UI

- Add filters, cards, distribution chart, table, drawer, radius controls, and export.

Exit criterion: Arizona/Tucson end-to-end scenario passes.

Epic 8: Accessibility, security, and documentation

- Complete accessible states and table alternative.
- Add security headers, rate limits, CSV safeguards, Terms, Privacy, and Data & Methods pages.
- Complete README, data dictionary, methods, and operations documents.

Exit criterion: full acceptance checklist passes from clean checkout.

23. Codex implementation rules

1. Keep Phase 1 read-only.
2. Do not substitute Google Places data for CMS or BTS source data.
3. Do not store Google Places search content beyond the active interaction.
4. Do not add schedule-aware GTFS features in this phase.
5. Prefer explicit SQL and tests for spatial calculations.
6. Never return an unbounded national raw-stop response.
7. Keep source snapshot IDs attached to metrics and exports.
8. Surface unresolved geocodes and source limitations.
9. Add tests with every functional change.
10. Update documentation whenever contracts, schemas, commands, or assumptions change.
11. Treat this document as authoritative unless a later written decision supersedes it.

24. Recommended first Codex prompt

Build the Phase 1 Dialysis & Transit Explorer described in docs/DESIGN.md.

Begin with Epic 1 only. Create the monorepo, Docker Compose environment, PostGIS database, Next.js TypeScript application, FastAPI service, health endpoints, test scaffolding, Makefile commands, .env.example, and a basic Google Maps page that can run with a configured API key. Do not implement data ingestion or analytics yet.

Before coding, summarize the files you will create and any assumptions. After coding, run the available lint, unit-test, build, and smoke-test commands; fix failures. Commit only files required for Epic 1 and update README.md with exact local setup instructions.

25. Future phases

- Agency-level GTFS routes, schedules, calendars, and shapes.
- Schedule-compatible dialysis-shift analysis.
- Walking-network and entrance-level routing.

- Census vehicle availability, disability, poverty, rurality, and vulnerability overlays.
- Transit frequency and weekend availability.
- Heat, shade, sidewalk, curb-ramp, and stop-amenity audits.
- Historical snapshots and change analysis.
- User-created study areas and saved projects.
- Statewide and metropolitan comparison reports.

These features must not be partially introduced into Phase 1 in ways that imply validated functional access.

26. Official references

1. [OpenAI Codex](#)
2. [Introducing the Codex app](#)
3. [Google Maps JavaScript API overview](#)
4. [Google Maps deck.gl Overlay View](#)
5. [Google Maps marker clustering example](#)
6. [Google Maps JavaScript API policies and attribution](#)
7. [Google Maps JavaScript API key setup](#)
8. [CMS Dialysis facilities topic](#)
9. [CMS Dialysis Facility - Listing by Facility](#)
10. [CMS Provider Data Catalog API documentation](#)
11. [BTS National Transit Map](#)
12. [BTS National Transit Map Stops FeatureServer](#)
13. [U.S. Census Geocoding Services API](#)
14. [U.S. Census batch address geocoder](#)

END OF SPECIFICATION